THE PRESIDENT'S OFFICE

REGIONAL ADMINISTRATIONS AND LOCAL GOVERNMENT

ARUSHA REGION

FORM FOUR MOCK EXAMINATION

BIOLOGY 2A

MARKING SCHEME

1. a)

Test for	Procedures	Observations	Inference
Starch	To 2cm3 of sample solution of l and k few drops of iodine solution was added	In both sample k and l blue – black coloration observed	Stach present
Reducing sugar	to 2cm3 of a sample solution of L . 2cm3 of Benedict solution was added and boiled for few minutes	series of colour changes was then observed from blue to green yellow, orange and finally brick red coloration in sample K and no series of color in sample L	This shows that reducing sugar was present in the food sample k and absent in food sample l
Non reducing sugar	To 2cm^3 of a food sample in a test tube K and L . 1cm^3 of Hcl was added to each sample and mixture was then boiled. NaoH was then added to each sample ($NaHCO3$) mixture. Equal volume of	The series of color change was then observed from blue yellow, orange and finally brick red coloration in each sample K and L	Non reducing sugar was present

	benedict's solution was added into a mixture and boiled again.		
Protein	To 2cm^{3}of K and L sample in a test tube a little solution of NaoH was added , followed by solution of copper II sulphate drop wise	There was no color change observed in both test tube.(retained colour)	Protein was absent
Lipids	to 2cm3of a sample K and L little amount of water was placed followed by addition few drops of suddan III dye solution and the mixture was shaken after 5mitutes @ 01	There was no any food droplets stapped by the Suddan iii solution at the upper part (top) @001/2	Lipids was absent @001/2

b) (b)(i) The nature of stored food substances in each storage orange

• In specimen K (sweet potato)

- Starch found in sweet potato remain as it is (is stored as starch) (01marks)

- Both reducing and non-reducing sugar found in potato are stored in form of starch since the storage of glucose in plants is starch (01 marks)

• In ginger (Specimen L)

- Starch found in ginger stored as starch (it remains as it was) (01marks)

- Reducing sugar and non-reducing sugar are both stored as starch since the storage of glucose in plants is starch (01 mark)

(ii) The plant part from which each storage organ develops

- For sweet potato (specimen K) was root tuber (01 marks)
- For ginger (specimen L) is rhizome (Modified plant stem) (01 marks)

(c) The role of Hcl in food test

- Dilute Hcl convert non- reducing sugar into reducing sugar. The sucrose is converted into glucose and fructose. (02 marks)

The role of NaoH

- To neutralize the acid

- To bring medium for proper working of benedict solution. (02 marks)

(d) The enzymes responsible is salivary amylase and the end product is Glucose (02 marks)

(e) (i) Heating is necessary for any benedict test because (02 marks)

- To increase activation energy for the reaction to takes place

- To increases vibration of molecules that causes collision

(ii) Production of Energy (01 mark)

2. (a)

A = Millipede

B = Centipede

C = Lizard

D = Beetle

E= Cockroach

F= Grasshopper

G = Bread mould (½ x6 = 03 marks)

b)	SPECIMEN	KINGDOM	PHYLUM	CLASS
A	Millipede	Animalia	Anthropoda	Diplopoda
B	Centipede	Animalia	Anthropoda	Chilopoda
C	Lizard	Animalia	Chordata	Reptilia
D	Beetle	Animalia	Anthropoda	Insecta
E	Cockroach	Animalia	Anthropoda	Insecta

F	Grasshopper	Animalia	Anthropoda	Insecta

@001/2= 09 marks

c. Observable features of A and B

i. They have hard exoskeleton (01marks)

ii. They have one pair of antennae (01marks)

ii. Their bodies are divided into segment (01marks)

(d) Mode of nutrition

- The mode of nutrition is heterotrophic nutrient (01marks)

i) Herbivorous (01marks)

ii) Carnivorous (01marks)

(e) Nutrition - Saprophyte mode of nutrition (1.5 marks)

Habitat – Grow on dead organic matter (1.5 marks)

(f)Habitat of specimen A and B

i. Moist shed places (01marks)

ii. Moist shed place (dark) 01 marks)

(g) Economic importance of specimen E

i) Source of food (01 mark)

ii) Destruction of property eg clothes, timber (01 marks)

iii) Spread of diseases (01 marks)